

# Computer Architecture (EE-3009)

## Assignment\_01

Fall 2026

**Due Date:** Sunday, September 20, 2026

**Total Points: 50**

**Instruction:**

- You need to submit the **neat and clean handwritten scan copy** of the assignment **on classroom** before due date
- In case of plagiarism, **zero marks** will be awarded to the students.

**Qus\_01:**

Consider two processors P1 and P2, executing the same instruction set. Assume that, under ideal conditions, for the same input, a program running on P2 takes 25% less time but incurs 20% more CPI as compared to program running on P1. if the clock frequency of P1 is 1 GHz, determine the clock frequency of P2 in GHz.

**Qus\_02:**

Assume a data center upgrades its infrastructure by deploying new enterprise-grade NVMe SSDs. The manufacturer quotes a single drive MTTF of 2,000,000 hours. The data center operates a Warehouse-Scale Computer (WSC) containing 25,000 servers, where each individual server is configured with exactly 4 of these NVMe SSDs.

For this scenario, answer the following:

- What is the FIT (failures in time) for a single NVMe SSD?
- What is the Failure Rate for the entire WSC?
- What is the MTTF of the entire WSC?
- If the MTTF is reduced to 1,500,000 hours, what is the updated MTTF of the WSC?
- The improvement in reliability is \_\_\_\_x. (fill in the blank; e.g. 3x, 4x)

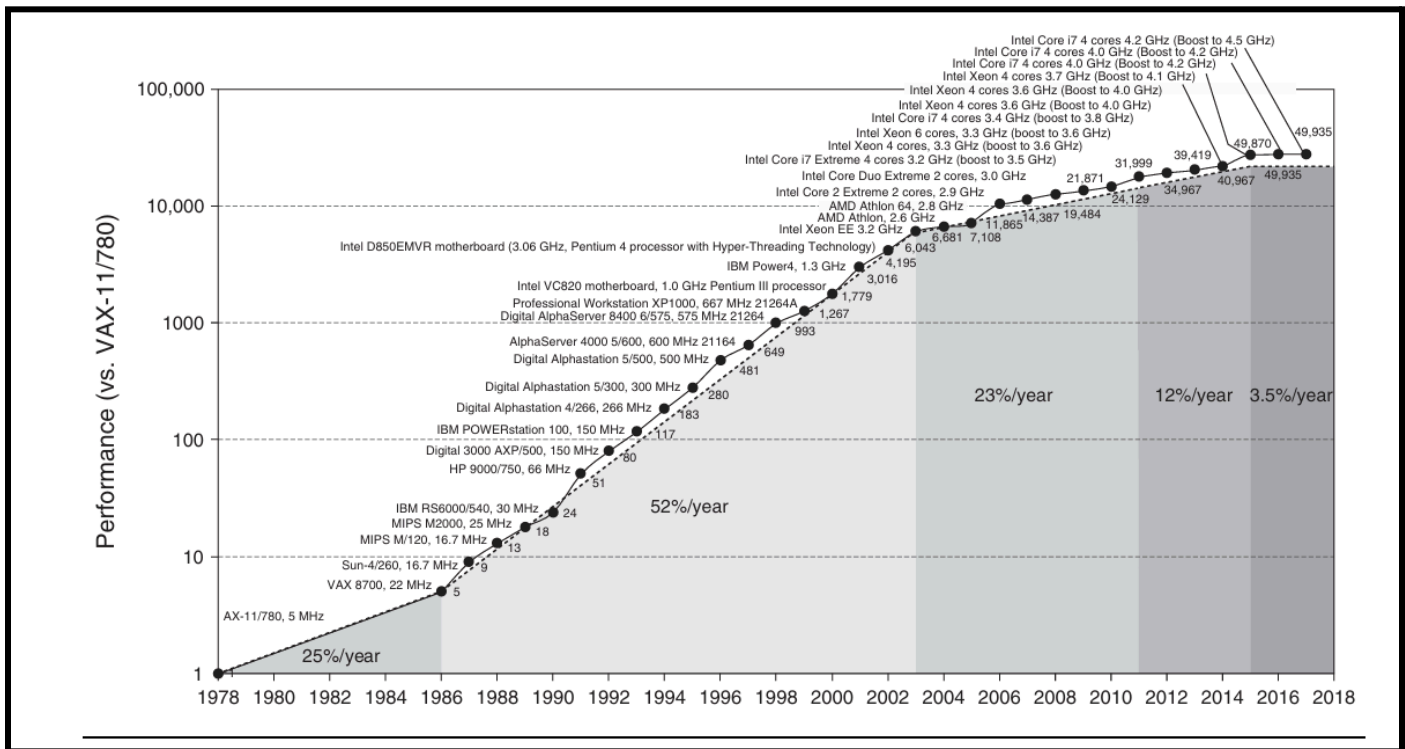
**Qus\_03:**

Consider a program that has the following instruction types. The frequency and CPI of each instruction type is given in the diagram below:

| Instruction Type    | Frequency | CPI |
|---------------------|-----------|-----|
| Integer Arithmetics | 40%       | 1   |
| Load                | 25%       | 3   |
| Store               | 25%       | 3   |
| FPSQRT              | 10%       | 10  |

- What is the Overall CPI for this program?
- What is the CPU Time of this program if the clock speed is 3 GHz and the Instruction Count is 30 billion.

**Qus\_04:** Consider Figure 1:



- Explain why processor performance improved significantly faster during the approximately 52% per year period compared with the earlier 25% per year period.
- Discuss the architectural and technological reasons that caused performance growth to slow down after approximately 2003.
- Explain how the breakdown of Dennard Scaling and the emergence of the Power Wall contributed to this slowdown.
- Based on your analysis, explain why simply increasing the clock frequency is no longer an effective method for continuously improving processor performance.

**Qus\_05:** Explain the purpose and importance of **SPEC** (Standard Performance Evaluation Corporation) benchmarks in evaluating computer system performance. In your answer, discuss:

- A. Why standardized benchmarks are needed to compare different processors.
- B. What is meant by SPECint and SPECfp.
- C. How SPEC benchmarks provide a fair comparison between computer systems.
- D. Why using a suite of different benchmarks is generally better than evaluating a processor using a single program.
- E. Explain what a higher SPEC performance score indicates about a processor.
- F. Also discuss one limitation of using benchmarks for measuring overall computer performance.